

# **OPNsense Firewall Configuration Walkthrough**

*Firewall Rule Configuration and Verification Process*

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## Introduction

This lab focuses on the installation and initial configuration of OPNsense in a virtual environment. It demonstrates the complete setup process, including installing OPNsense using the ZFS file system, assigning WAN and LAN interfaces, configuring the LAN IP address, setting up the Web GUI, completing the initial configuration wizard, and customizing the dashboard. These steps establish a functional OPNsense firewall environment that is ready for firewall rule configuration, traffic filtering, and security testing in the subsequent lab.

### 1. Installer Boot and Task Selection

The OPNsense installation begins with the FreeBSD based installer running inside the VirtualBox virtual machine. The main menu titled "OPNsense 26.1" lists the available installation tasks, including Install (ZFS), Install (UFS), Other Modes, Import Config, Password Reset, Force Reboot and Force Halt. The Install (ZFS) option, described as ZFS GPT/UEFI Hybrid, is selected to proceed with a ZFS based installation.

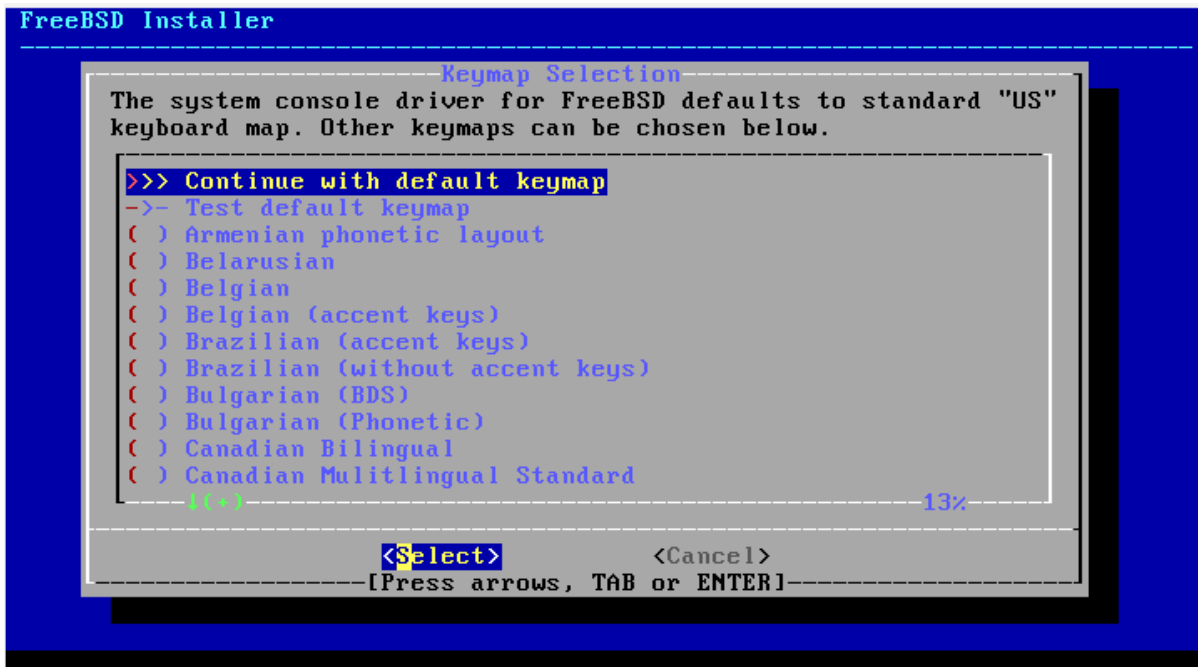


Figure 1: OPNsense Installer Main Menu

## 2. Keymap Selection

After choosing the installation type, the installer displays the Keymap Selection screen. The default keyboard map for the system console is standard US, and the option Continue with default keymap is selected to keep the default layout before moving to the next step.



Figure 2: Keymap Selection Screen

### 3. ZFS Virtual Device Type

The ZFS Configuration screen asks for the Virtual Device type to be used for the pool. The available options are stripe, mirror, raid10, raidz1, raidz2 and raidz3. The stripe option, described as Stripe - No Redundancy, is selected since the lab uses a single virtual disk.

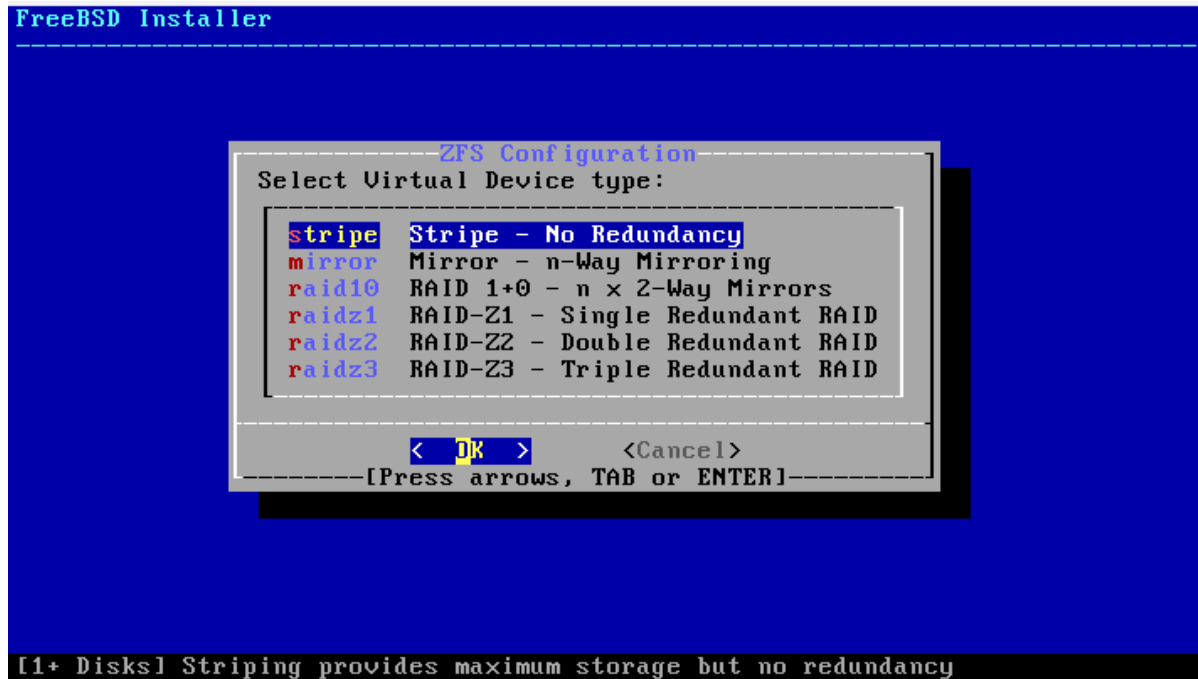


Figure 3: ZFS Virtual Device Type Selection - Stripe

## 4. ZFS Disk Selection

The installer then lists the available disks for creating the zpool. Only one disk, ada0, labelled VBOX HARDDISK, is present in the virtual machine. This disk is selected before proceeding with OK.

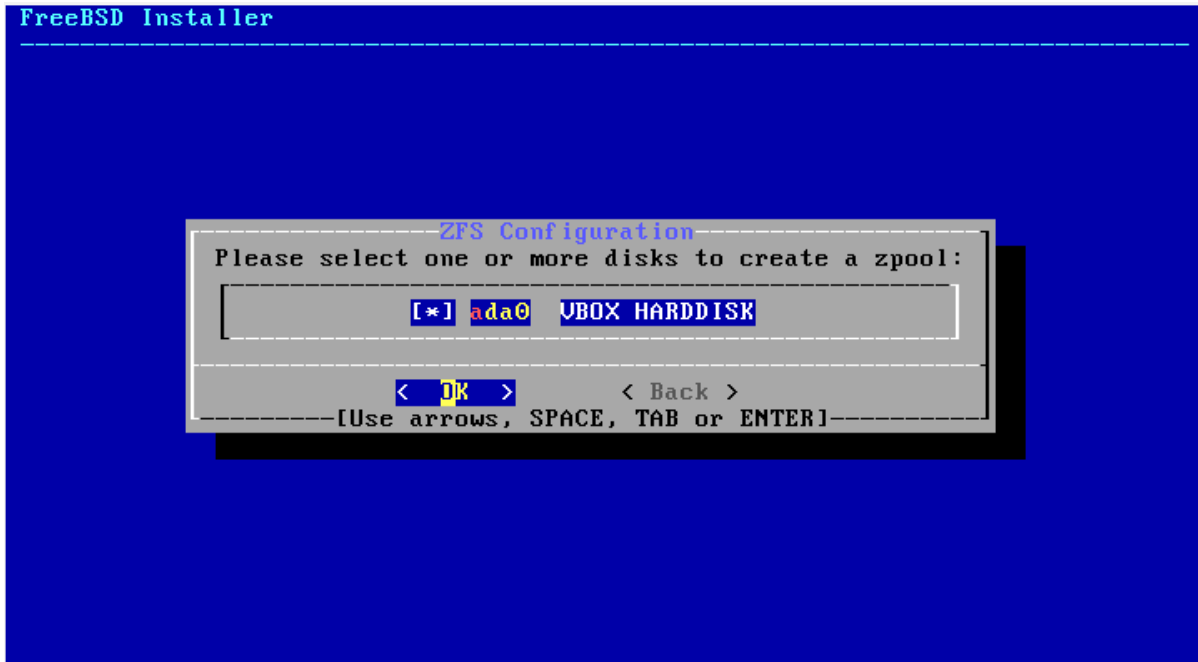


Figure 4: ZFS Disk Selection - ada0 VBOX HARDDISK

## 5. Confirm Disk Erase

Before writing the ZFS pool, the installer shows a last chance warning confirming that the current contents of disk ada0 will be destroyed. Selecting YES confirms the disk erase and allows the installation to continue.

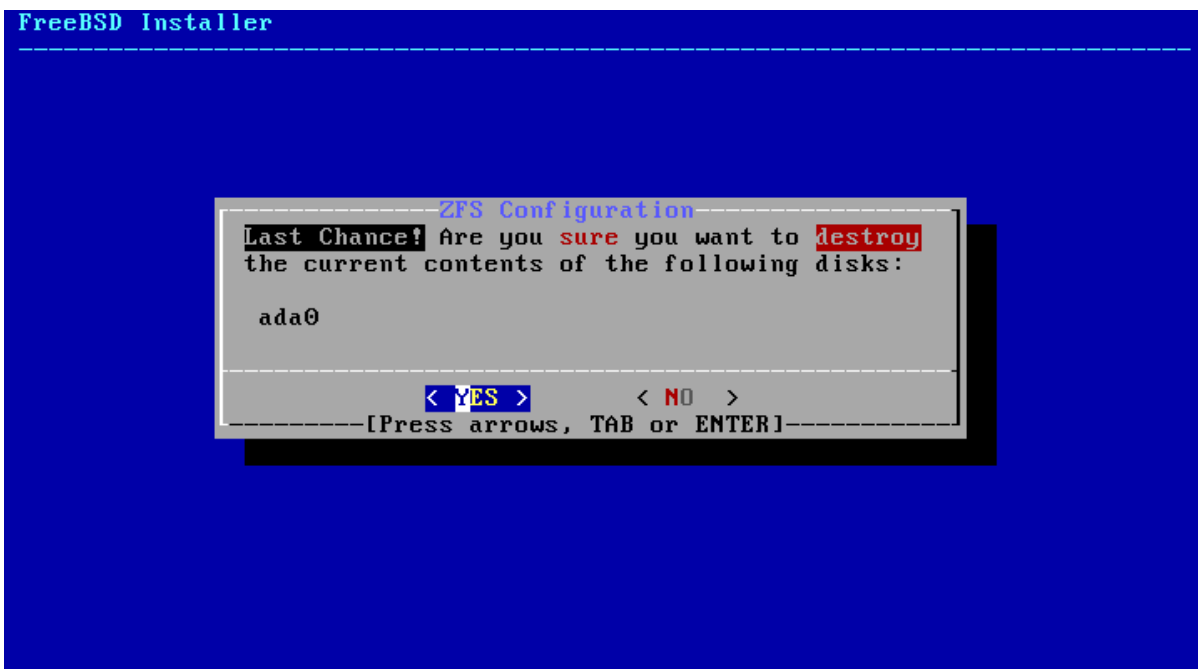


Figure 5: Confirm Disk Erase Warning for ada0

## 6. Installation Progress

Once the disk is confirmed, the installer begins copying the OPNsense system files. The Installation Progress window shows the current task, Cloning current system, along with an individual progress bar at 30 percent and an Overall Progress bar at 24 percent.

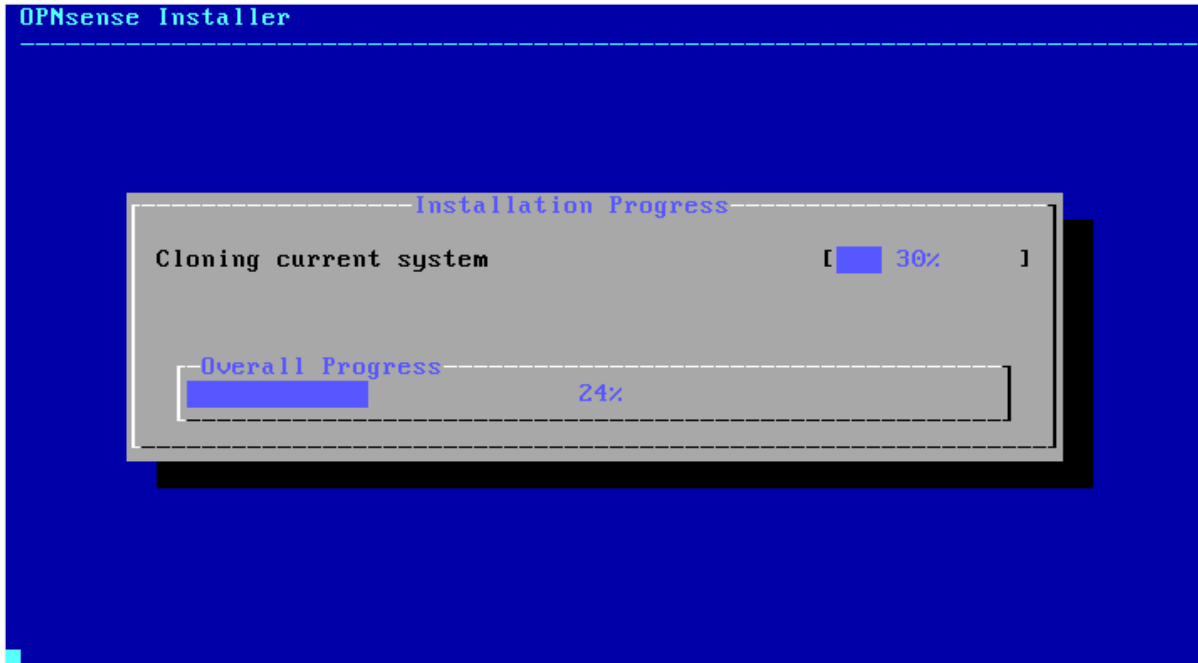


Figure 6: OPNsense Installation Progress

## 7. Final Configuration

After the files are copied, the Final Configuration screen appears indicating that the setup is nearly complete. Two options are available, Root Password to change the root password and Complete Install to confirm and exit. Complete Install is selected to finish the setup.

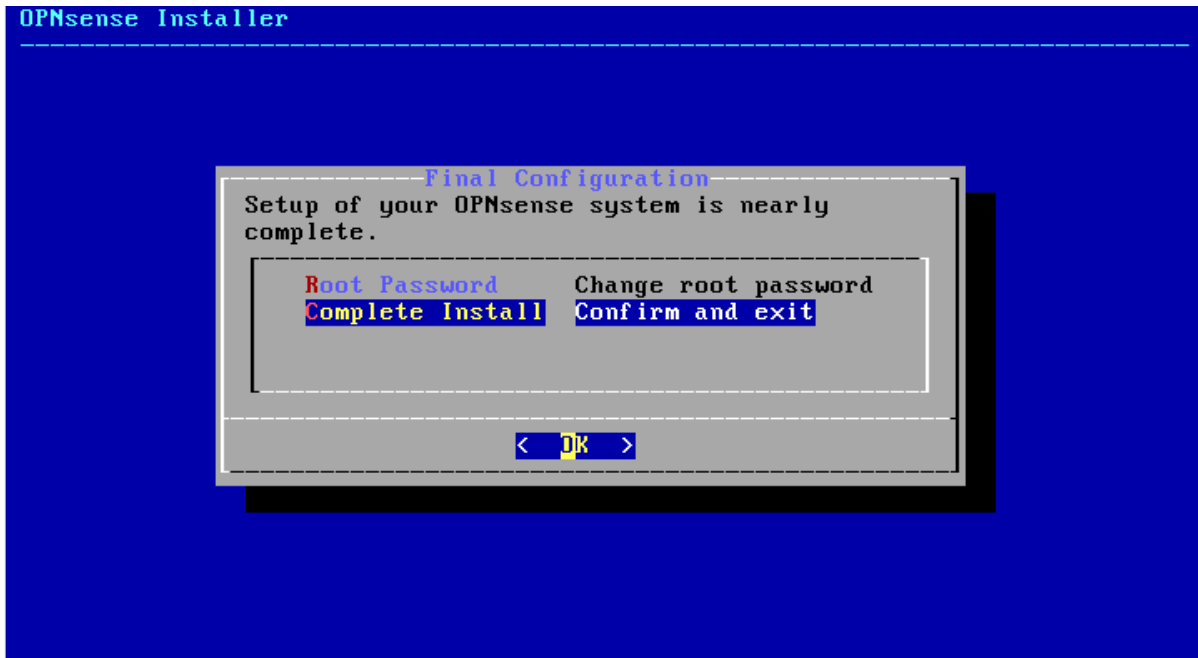


Figure 7: Final Configuration - Complete Install



## 8. Installation Complete

The Installation Complete screen notes that the system may boot back into the installation media if it is not ejected properly. The available options are Reboot now and Halt now, described as Power down system. Halt now is selected so the installation media can be removed before the virtual machine is powered back on.

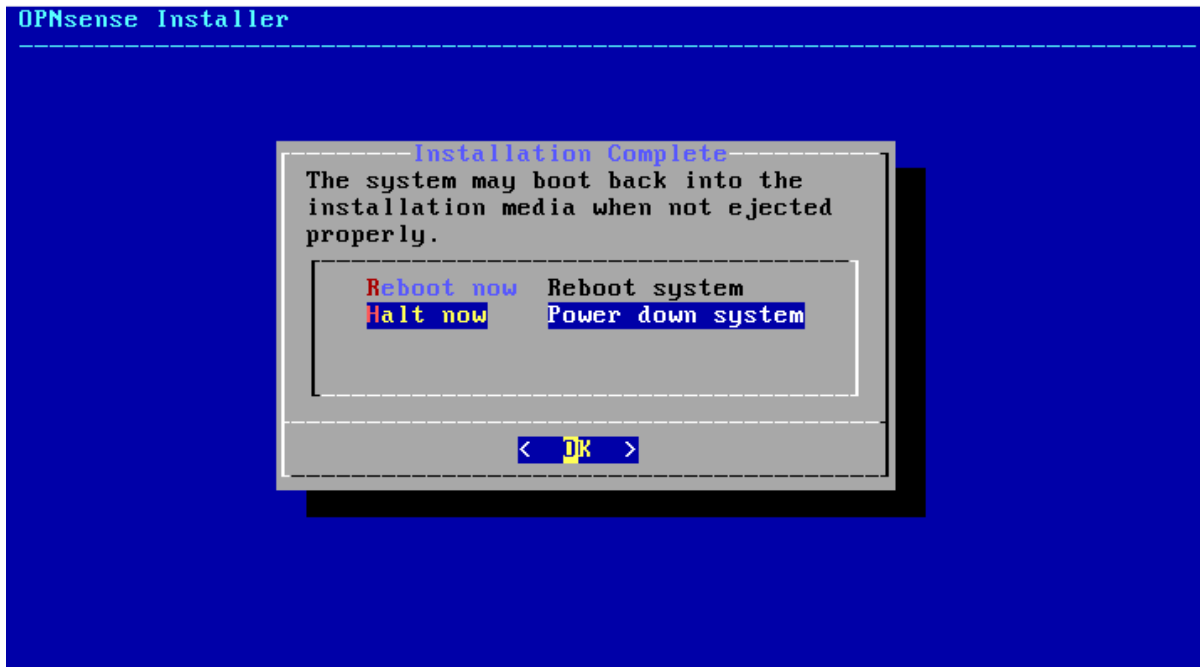


Figure 8: Installation Complete - Power Down System

## 9. Interface Assignment - WAN and LAN

On first boot, the console menu is used to assign interfaces. LAGGs and VLANs are skipped, and the valid interfaces detected are em0 and em1. The WAN interface is set to em0 and the LAN interface to em1, enabling full Firewalling and NAT mode.

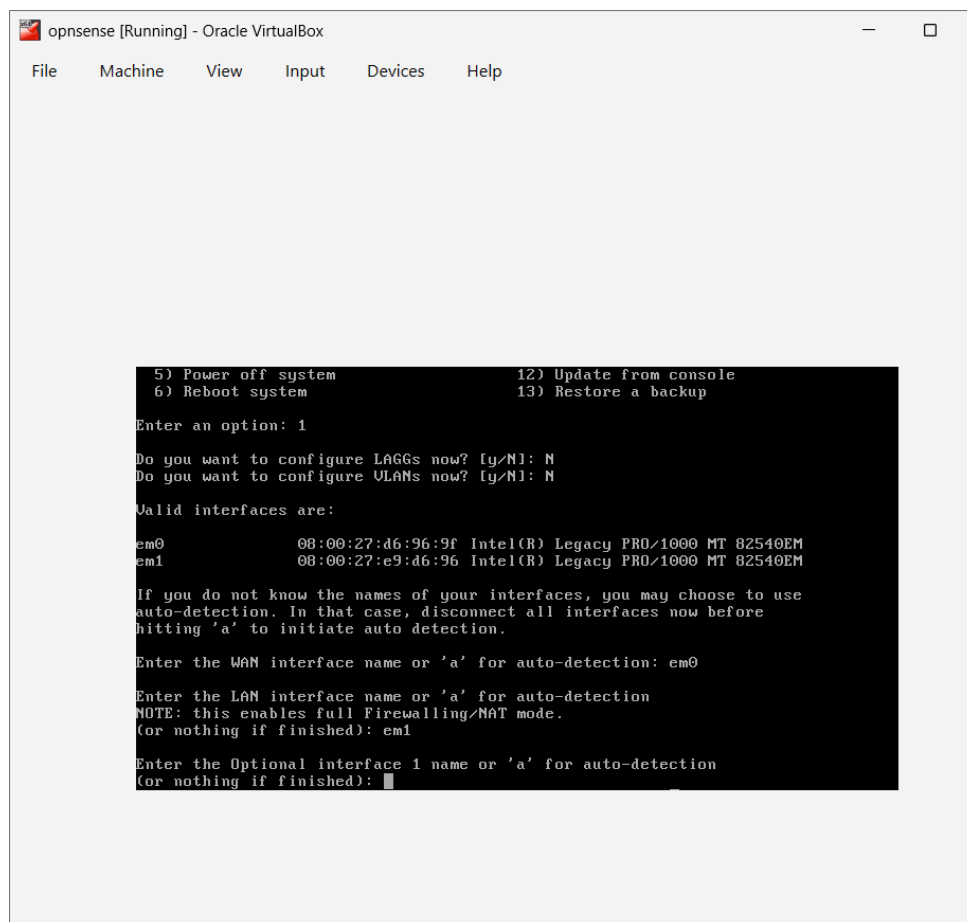


Figure 9: Interface Assignment - WAN and LAN Selection

## 10. Applying the Interface Configuration

With no optional interfaces added, the summary confirms WAN -> em0 and LAN -> em1, and entering y applies it. The console then writes the configuration and starts each service in turn, including the LAN and WAN interfaces, routes, gateway monitor, firewall, Dnsmasq, NTP and Unbound DNS.

```
Enter the Optional interface 1 name or 'a' for auto-detection
(or nothing if finished):

The interfaces will be assigned as follows:

WAN  -> em0
LAN  -> em1

Do you want to proceed? [y/N]: y

Writing configuration...done.
Configuring loopback interface...done.
Configuring LAN interface...done.
Configuring WAN interface...done.
Setting up routes...done.
Setting up gateway monitor...done.
Configuring firewall.....done.
Starting Dnsmasq...done.
Starting NTP service...done.
Starting Unbound DNS...█
```

Figure 10: Interface Configuration Applied and Services Starting

## 11. OPNsense Console Menu

Once the services finish, the status banner shows LAN (em1) at 192.168.1.1/24 and WAN (em0) obtaining an address via DHCP4/DHCP6, along with the HTTPS certificate fingerprint. From the console menu, option 2, Set interface IP address, is selected to change the LAN address.

```

Configuring firewall.....done.
Starting Dnsmasq...done.
Starting NTP service...done.
Starting Unbound DNS...done.
Starting web GUI...done.
Syncing OpenVPN settings...done.

*** OPNsense.internal: OPNsense 26.1.6_2 (amd64) ***

LAN (em1)      -> v4: 192.168.1.1/24
WAN (em0)      -> v4/DHCP4: 192.168.177.58/24
                v6/DHCP6: 2401:4900:16ac:b163:a00:27ff:fed6:969f/64

HTTPS: SHA256 F2 37 0D 4B 2D B2 2B F2 45 8D 4B 2A 15 45 18 1C
          66 0B 44 36 F5 6F F1 5C 20 8D 2C F9 66 89 80 F9

0) Logout              7) Ping host
1) Assign interfaces   8) Shell
2) Set interface IP address  9) pfTop
3) Reset the root password 10) Firewall log
4) Reset to factory defaults 11) Reload all services
5) Power off system       12) Update from console
6) Reboot system          13) Restore a backup

Enter an option: 2

```

Figure 11: OPNsense Console Menu - Set Interface IP Address

## 12. VirtualBox Host-Only Adapter Configuration

Before the LAN address is changed on OPNsense, the VirtualBox Network Manager is used to configure the Host-only network that the LAN interface connects to. VirtualBox Host-Only Ethernet Adapter #2 is configured manually with IPv4 Address 192.168.230.1 and IPv4 Network Mask 255.255.255.0, so that it can communicate with the new LAN subnet.



Figure 12: VirtualBox Host-Only Adapter Configuration

### 13. Configuring the LAN IP Address

Back in the OPNsense console, interface 1, LAN (em1 - static, idassoc6), is selected for configuration. DHCP is not used for the LAN IPv4 address, and the new LAN IPv4 address is entered as:

```
> 192.168.230.2
```

The subnet bit count is entered as 24, and no upstream gateway is set since this is a LAN interface.

```
Available interfaces:

1 - LAN (em1 - static, idassoc6)
2 - WAN (em0 - dhcp, dhcp6)

Enter the number of the interface to configure: 1

Configure IPv4 address LAN interface via DHCP? [y/N] N

Enter the new LAN IPv4 address. Press <ENTER> for none:
> 192.168.230.2

Subnet masks are entered as bit counts (like CIDR notation).
e.g. 255.255.255.0 = 24
     255.255.0.0   = 16
     255.0.0.0     = 8

Enter the new LAN IPv4 subnet bit count (1 to 32):
> 24

For a WAN, enter the new LAN IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
> 
```

Figure 13: Configuring LAN IPv4 Address 192.168.230.2/24

## 14. LAN Web GUI Settings

IPv6 address configuration through WAN tracking and DHCP6 is both declined for the LAN interface, and no IPv6 address is set. The DHCP server on LAN is left disabled, and the web GUI protocol is kept as HTTPS instead of HTTP. A new self-signed web GUI certificate is generated, and the web GUI access defaults are restored so the interface can be reached from the browser.

```
Subnet masks are entered as bit counts (like CIDR notation).
e.g. 255.255.255.0 = 24
     255.255.0.0   = 16
     255.0.0.0     = 8

Enter the new LAN IPv4 subnet bit count (1 to 32):
> 24

For a WAN, enter the new LAN IPv4 upstream gateway address.
For a LAN, press <ENTER> for none:
>

Configure IPv6 address LAN interface via WAN tracking? [Y/n] n
Configure IPv6 address LAN interface via DHCP6? [y/N] N

Enter the new LAN IPv6 address. Press <ENTER> for none:
>

Do you want to enable the DHCP server on LAN? [y/N] N

Do you want to change the web GUI protocol from HTTPS to HTTP? [y/N] N
Do you want to generate a new self-signed web GUI certificate? [y/N] y
Restore web GUI access defaults? [y/N] y
```

Figure 14: LAN Web GUI Certificate and Access Settings

## 15. OPNsense Web GUI Login

With the new LAN address applied, the OPNsense web interface is opened in the browser at <https://192.168.230.2>. The login page displays the OPNsense logo with the tagline Securing networks made easy. The default credentials, username root and password opnsense, are entered to log in.

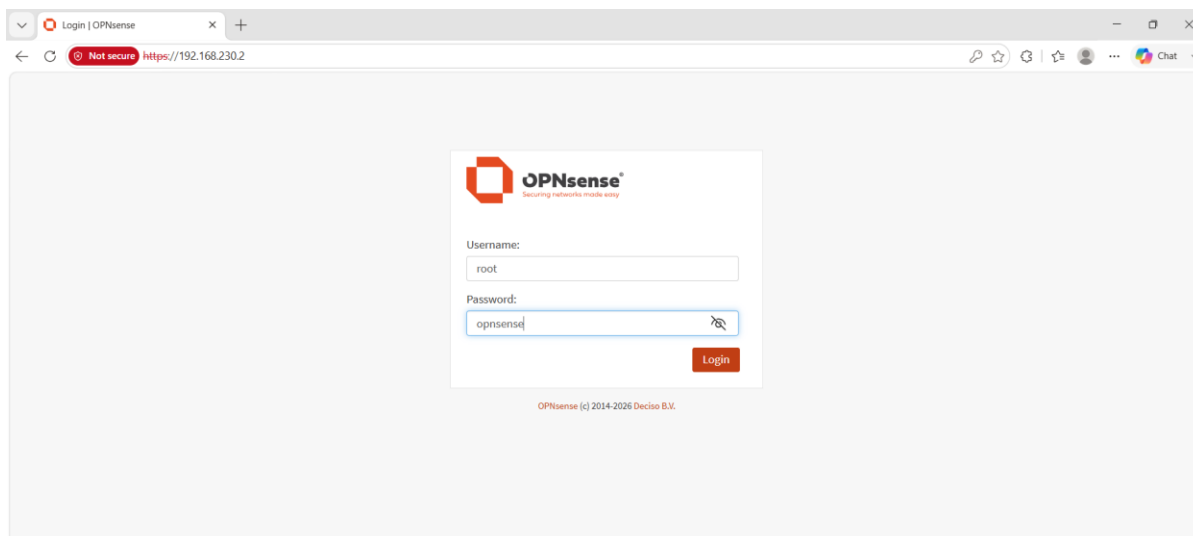


Figure 15: OPNsense Web GUI Login Page

## 16. Configuration Wizard - General Information

After logging in, the System: Configuration: Wizard opens on the General Information tab. Hostname is set to OPNsense, Domain to internal, Language to English and Timezone to Asia/Kolkata. DNS Servers lists 8.8.8.8 with Override DNS and Enable Resolver checked, while DNSSEC options stay unchecked.

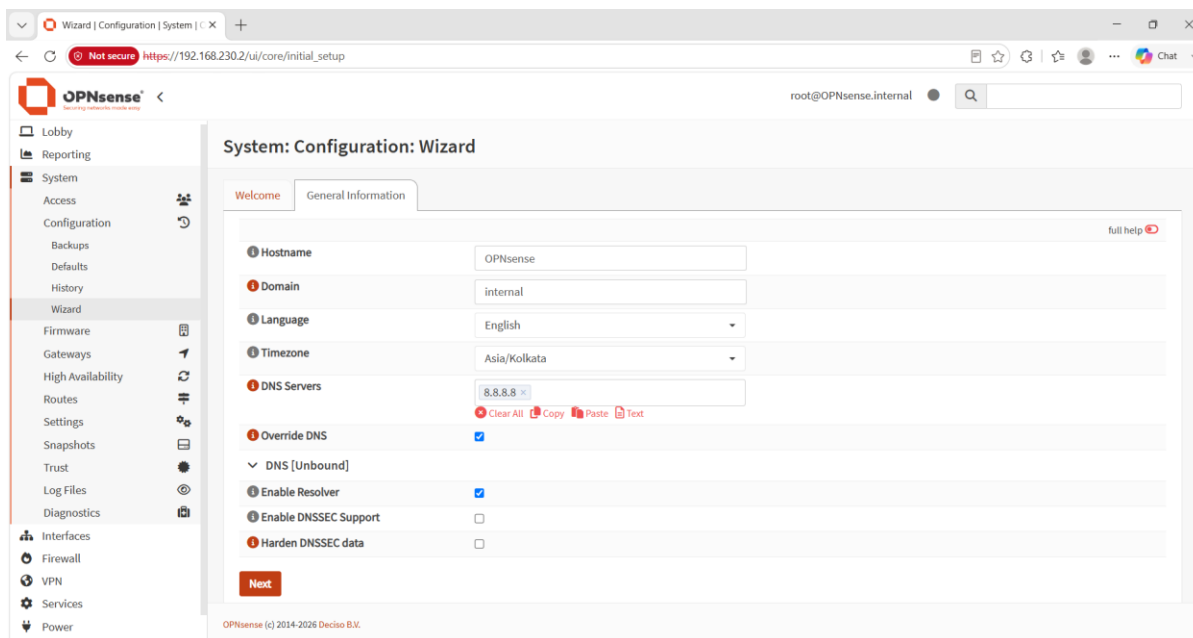


Figure 16: Configuration Wizard - General Information



## 17. Configuration Wizard - Finish

The wizard proceeds through the Network [WAN], Network [LAN], Deployment type and Set initial password tabs to the final Finish tab. The page states that this is the last step in the wizard and clicking Apply will reconfigure the firewall. Apply is clicked to complete the initial setup wizard.

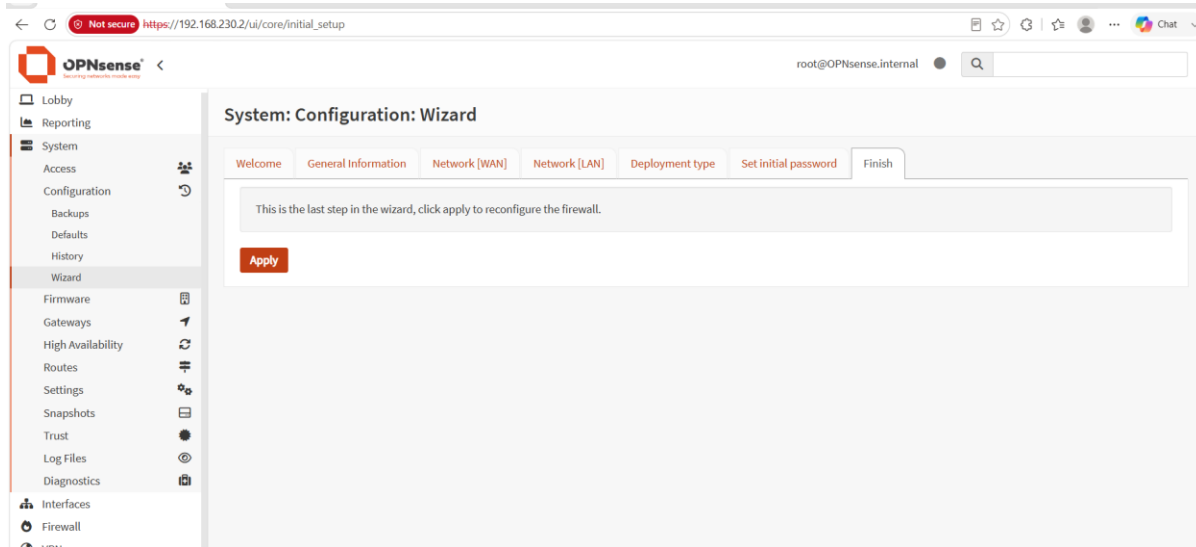


Figure 17: Configuration Wizard - Finish and Apply

## 18. Opening the Add Widget Menu

On the Lobby: Dashboard page, the Add Widget icon in the top right corner of the dashboard is used to open the widget selection menu. The dashboard already shows a Firewall widget as a donut chart with categories such as let out anything from firewall host and IPv6 RFC4890 requirement, along with a Traffic Graph widget showing Traffic In.

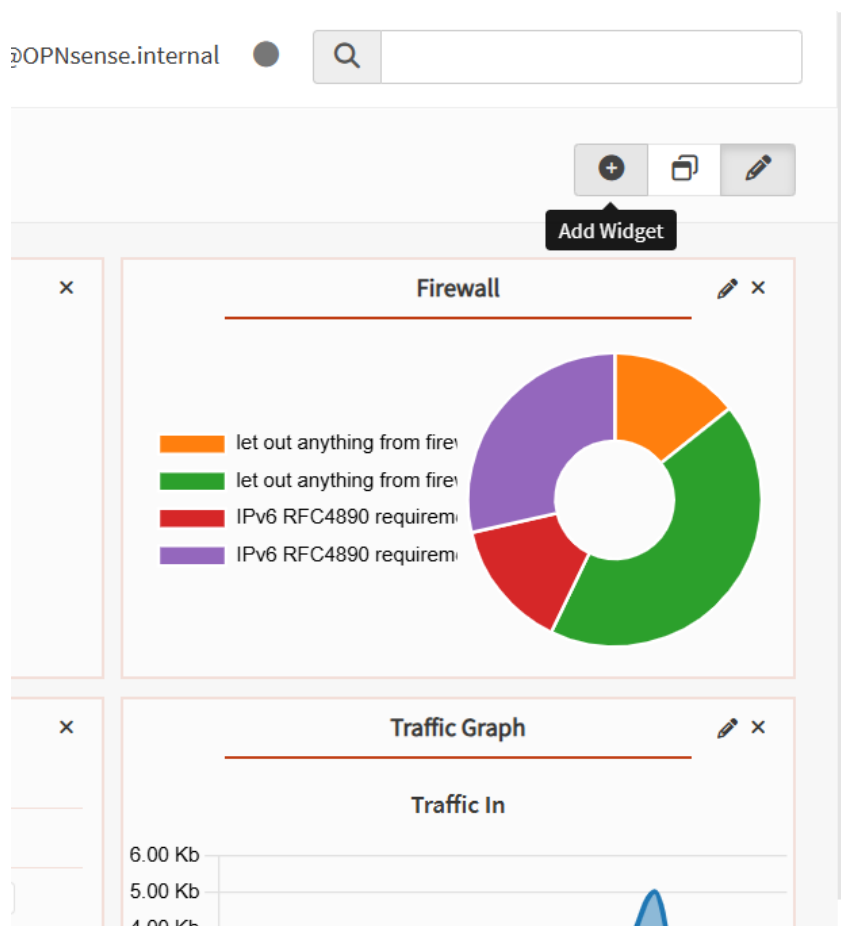


Figure 18: Dashboard - Opening the Add Widget Menu

## 19. Adding the Interfaces Widget

In the Add Widget dialog, the dropdown lists all available dashboard widgets, including CARP Status, Certificates, Dnsmasq DHCP, Firewall States, Interfaces, IPsec Leases, IPsec Tunnels, Live Log, Mbuf, Monit Status, OpenVPN Client Connections, OpenVPN Server Connections, Picture, Swap, Thermal Sensors and Wireguard. The Interfaces option is selected, ready to be added to the dashboard alongside the existing System Information, Firewall, Services and Traffic Graph widgets.

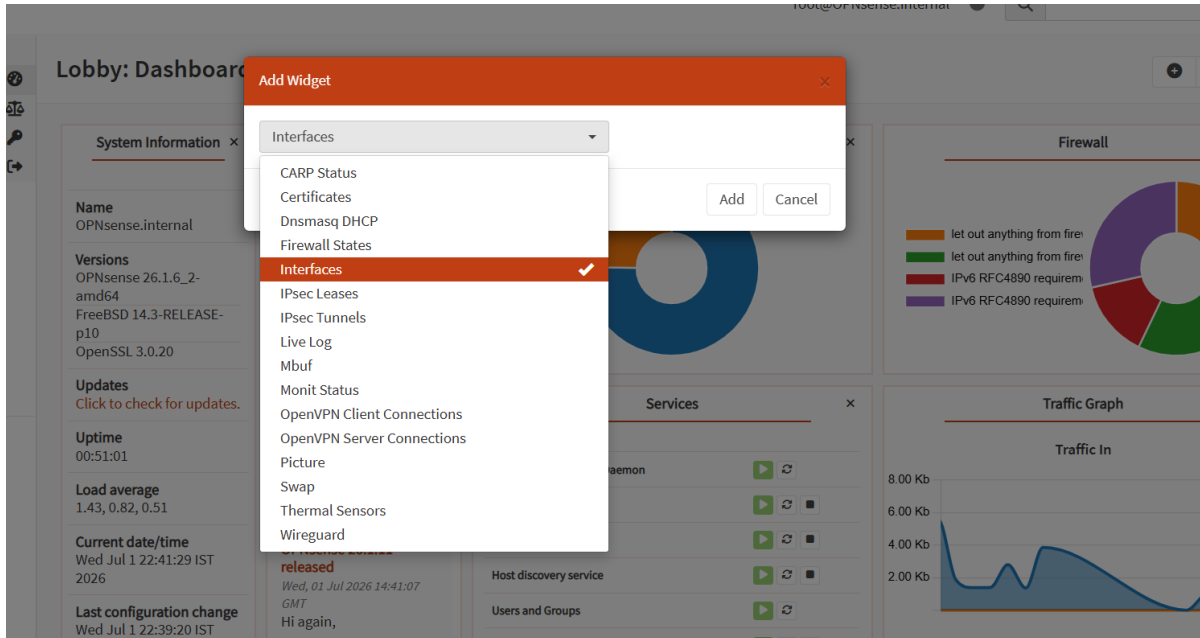


Figure 19: Add Widget Dialog - Interfaces Selected

## Conclusion

This walkthrough covered the complete process of installing and performing the initial configuration of OPNsense in a virtual lab environment. The installation was completed using the ZFS file system, followed by interface assignment, LAN IP configuration, VirtualBox Host-Only Adapter setup, and secure Web GUI access. The initial configuration wizard was then completed, the dashboard was customized by adding the required widget, and the LAN interface was renamed to Homelab. These steps established a fully functional OPNsense environment, providing a stable foundation for the firewall configuration, rule creation, and traffic filtering tasks performed in the next lab.